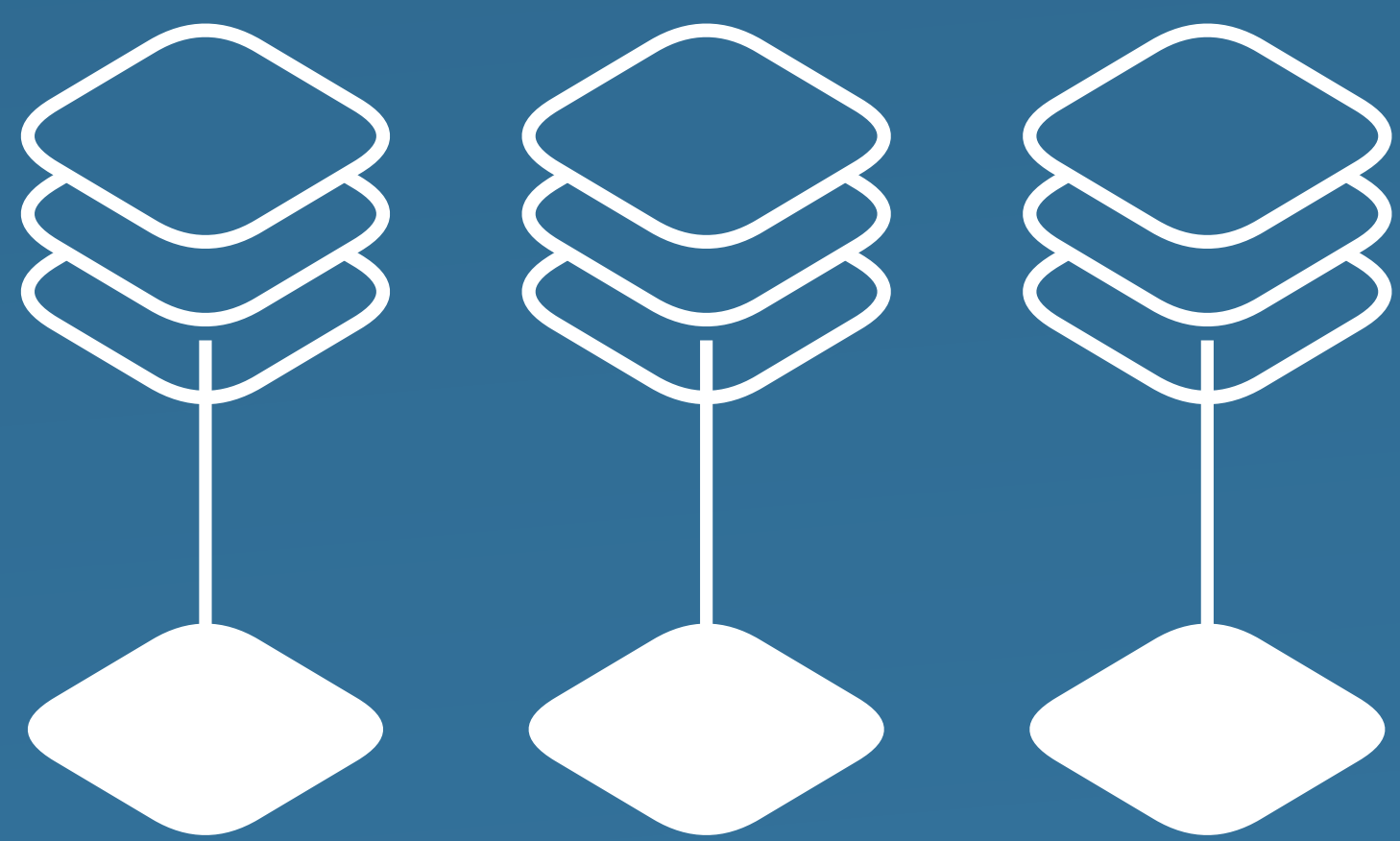




Which layer of a frozen vision foundation model should you probe?

None alone.

For dense prediction, try **AMDF**: it fuses **all** layers, for **every** patch.



On DINOv3-L, each NYUv2 task peaks at a different depth

**≈22×**

fewer params than concatenating layers (dense tasks)



Let's talk!

Open to research collaborations

vision foundation models · representation learning · probing & interpretability · spatial understanding



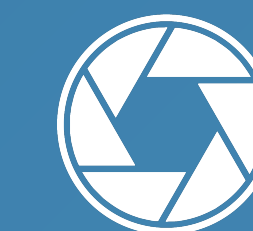
turhancankargin.me

# AMDF: Holistic Probing of Vision Foundation Models with Adaptive Multi-Layer Dense Fusion

Turhan Can Kargin<sup>1,2</sup> · Bill Psomas<sup>3,4</sup> · Paul Wagner<sup>5</sup> · Giorgos Tolias<sup>3,4</sup> · Bartosz Zieliński<sup>1,2</sup> · Marcin Przewięźlikowski<sup>1,2</sup>

<sup>1</sup> Jagiellonian University in Kraków <sup>2</sup> GMUM <sup>3</sup> Czech Technical University in Prague <sup>4</sup> VRG <sup>5</sup> RWTH Aachen University

Contact: turhancan.kargin@doctoral.uj.edu.pl



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## Tasks live at different depths

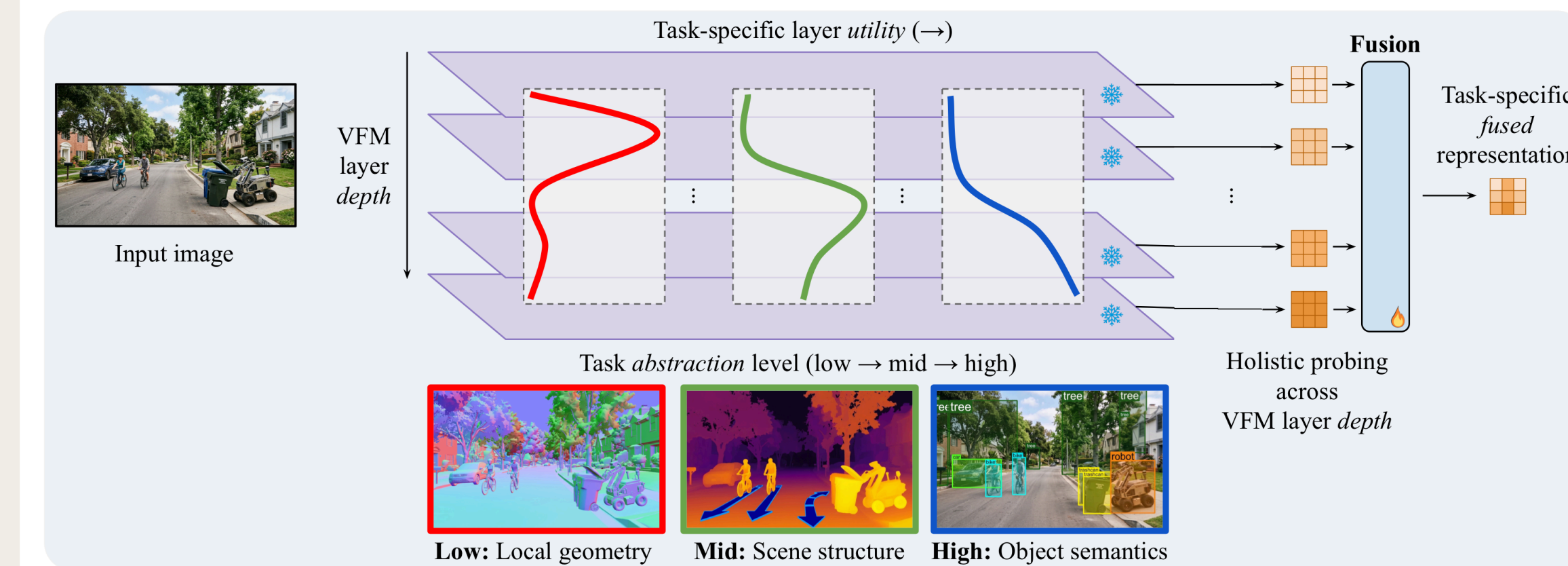


Fig. 1: Geometry peaks mid-depth, semantics at the end; AMDF fuses all layers per patch.

## #1: last layer ≠ always best

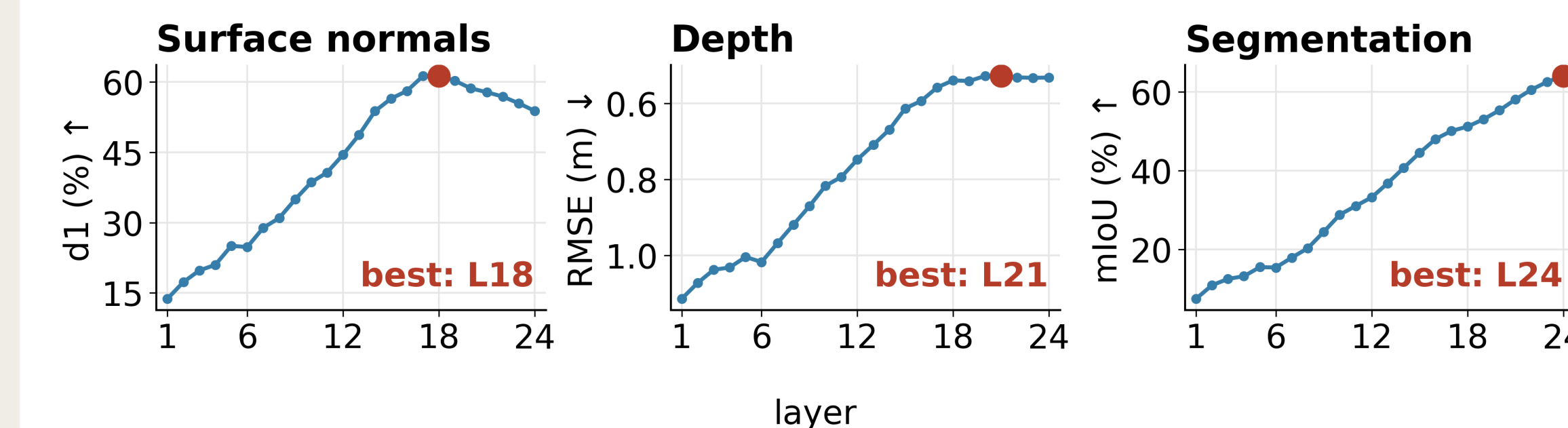


Fig. 2: Best single layer differs per task: ≈L18 (normals), ≈L21 (depth), L24 (segmentation); one seed per layer (DINOv3-L, NYUv2).

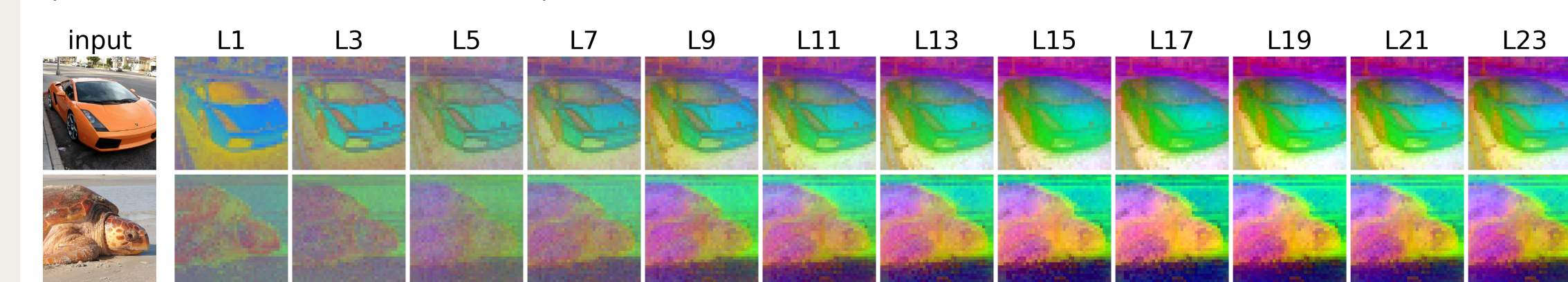


Fig. 3: MAE-L, ImageNet: fine detail early, object regions and parts (turtle shell vs head) deep.

## #2: all layers = costly

✗ Concat all layers: **26.2M params**, the same fixed weights for every patch.

✓ AMDF: **per-patch** layer choice, **≈1.2M params**.

## Take-home

- Useful features are **spread across depth**.
- AMDF: **2nd-best mean rank** with **≈22×** fewer params than Concat.
- AMDF beats every **fixed layer mix** by mean rank.

7 frozen VFMs

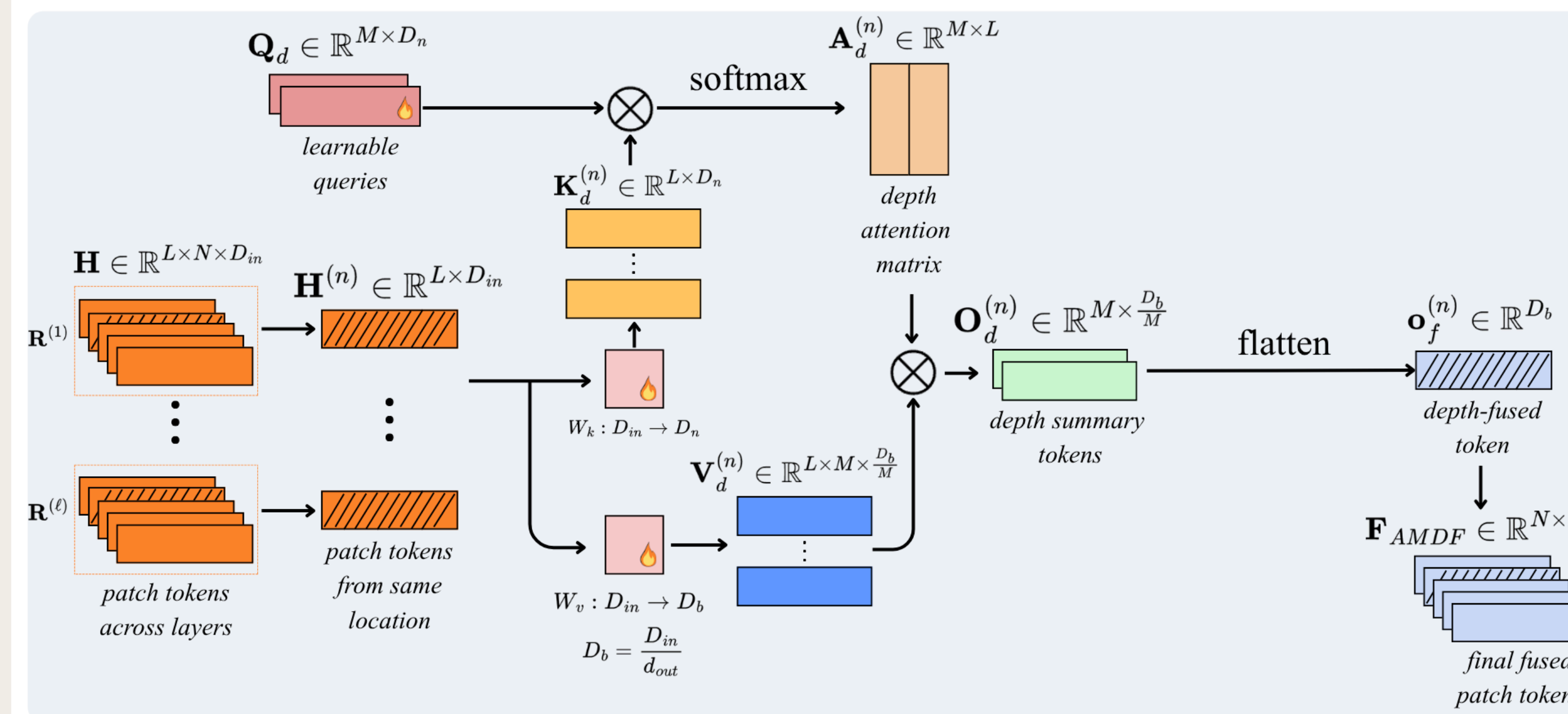
5 benchmarks

5 baselines

≤3 seeds

## Solution: AMDF — depth-aware fusion

- Per patch, learnable queries score all layers
- Softmax over depth → a task-specific layer mix
- Frozen backbone, light task heads



## Attention vs. layer utility (earlier variant)

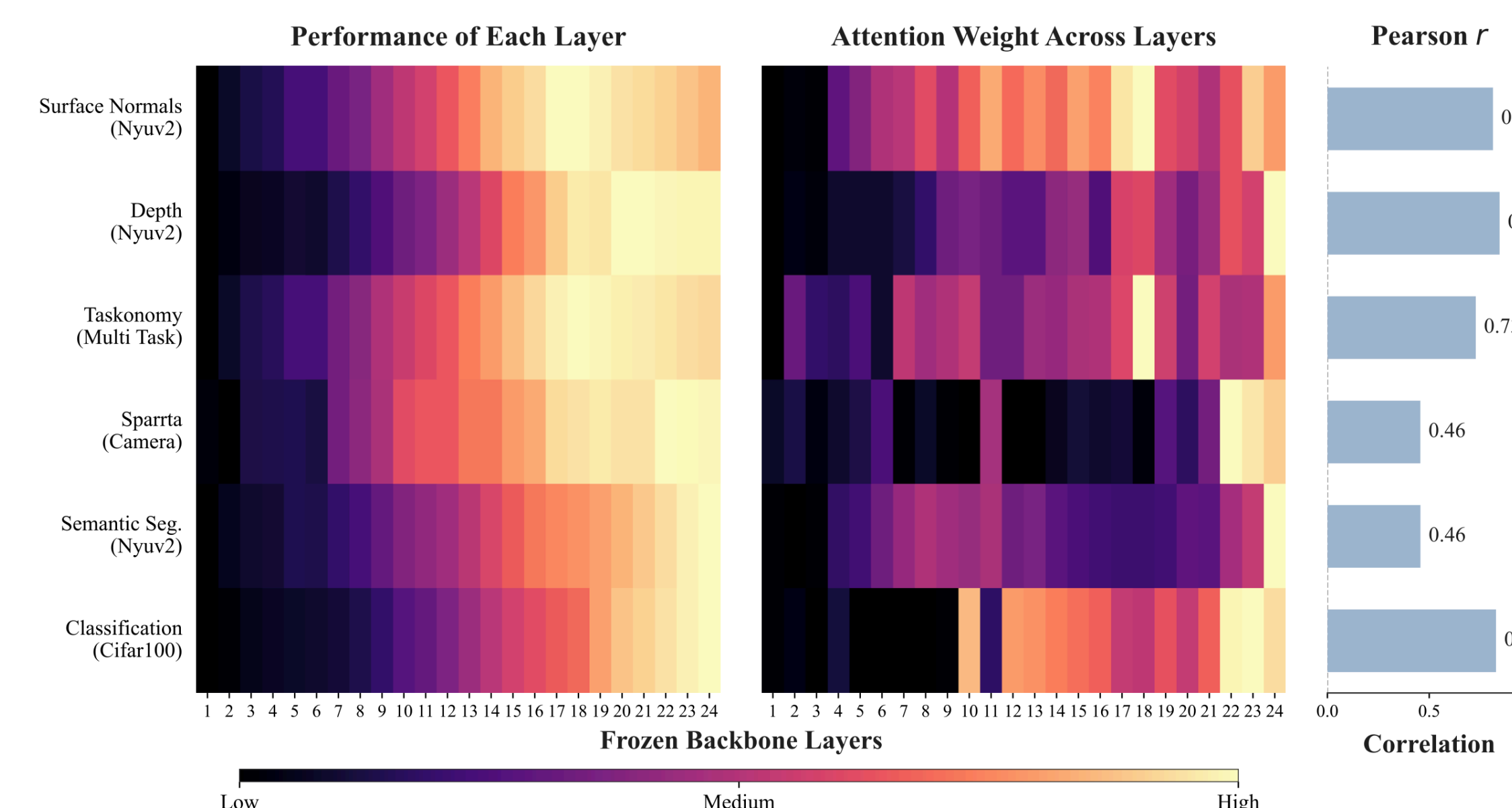


Fig. 4: Earlier AMDF variant (DINOv3-L): attention correlates with per-layer probe utility,  $r = 0.46\text{--}0.85$  across 6 tasks.

## What AMDF predicts

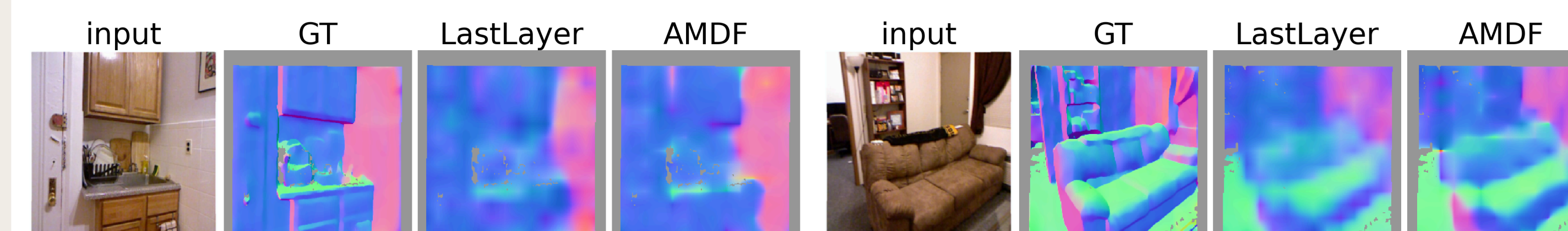


Fig. 5: NYUv2 surface normals, DINOv3-L (seed-8 retrain): AMDF recovers sharper geometry than the last-layer probe, though both are smoother than the ground truth.

## Per-layer queries: better mean on all 5 tasks

| Normals       | Taskonomy     | Segm.         | Depth         | Pose          |
|---------------|---------------|---------------|---------------|---------------|
| +1.62 d1      | −0.73 L1      | +0.16 mIoU    | +0.51 d1      | −2.37° rot.   |
| 7/7 backbones | 7/7 backbones | 6/7 backbones | 5/7 backbones | 5/7 backbones |

Paired gains vs. shared queries, mean over 7 backbones; Taskonomy L1 ×100; cost: +2% params.

## Results

**≈22×** fewer params than Concat  
**2nd** mean rank (1st on depth)  
**5×7** benchmarks × backbones

## Mean rank per benchmark

|                    | Normals       | Depth         | Segm.         | Taskonomy     | Pose          |
|--------------------|---------------|---------------|---------------|---------------|---------------|
| <b>AMDF (ours)</b> | 2.00          | <b>1.43 ★</b> | 1.86          | 2.00          | 1.93          |
| ConcatLayers       | <b>1.00 ★</b> | 1.71          | <b>1.57 ★</b> | <b>1.00 ★</b> | <b>1.21 ★</b> |
| WeightedLayers     | 4.14          | 4.00          | 4.14          | 3.29          | 3.89          |
| SpatialWeights     | 3.14          | 4.14          | 4.43          | 4.00          | 4.00          |
| LastLayer          | 5.43          | 4.14          | 3.86          | 5.29          | 4.96          |
| AvgLayer           | 5.29          | 5.57          | 5.14          | 5.43          | 5.00          |

Fig. 6: Mean rank per benchmark (★ best). Concat 1st on 4/5, AMDF 1st on depth.

## Accuracy vs. parameters

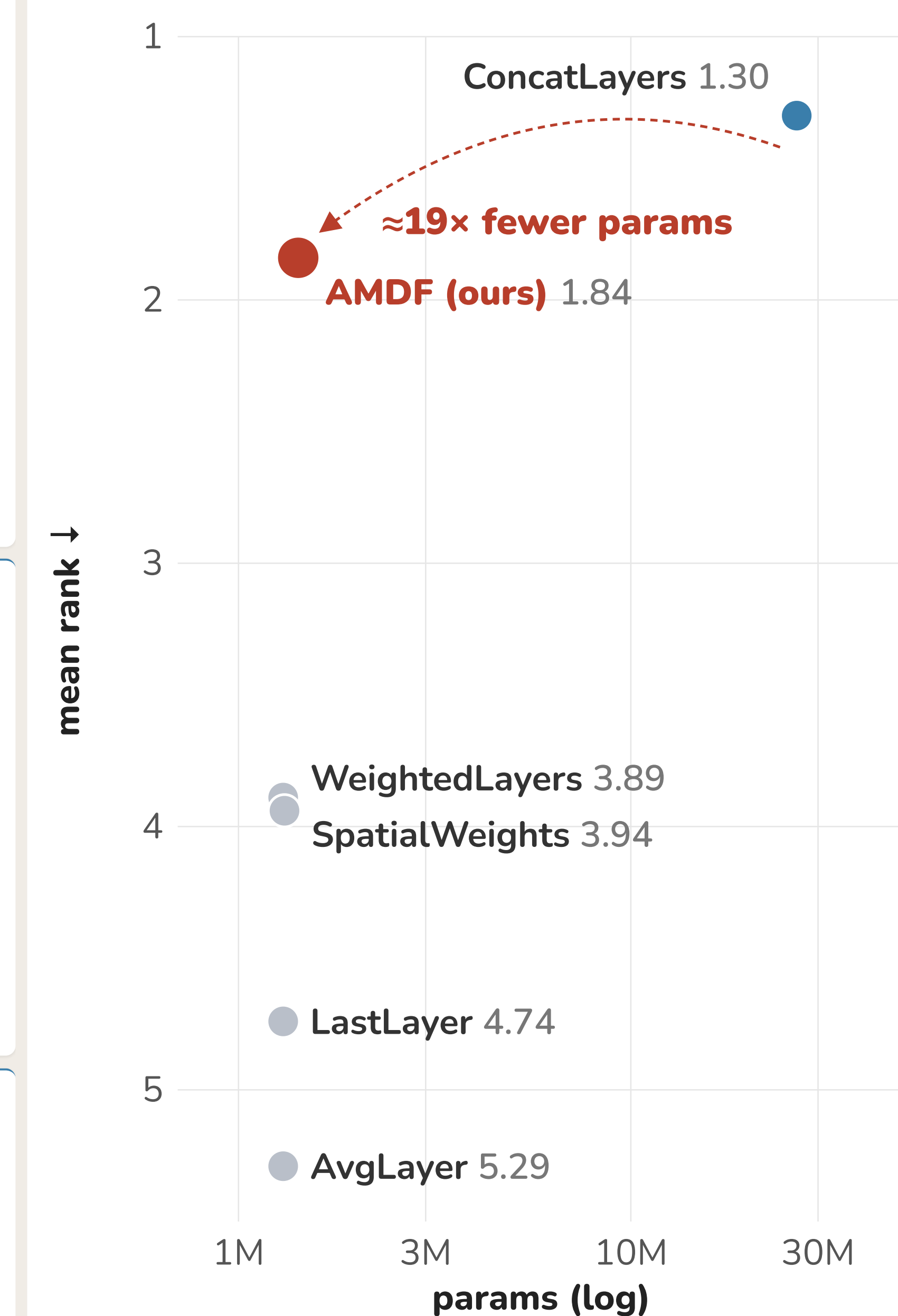


Fig. 7: ≈19× fewer params than Concat (5-task average, DINOv3-L).